CSE 311L (Database Management System)

LAB-Week 04 (Part A)

Displaying Data from Multiple Tables (Based on Company2.sql)

Topics:

- Obtaining Data from Multiple Tables
- Generating a Cartesian Product
- Retrieving Records with Equijoins
- Joining a Table to Itself
- Creating Joins with the ON Clause

EMPLOYEES

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID
100	King	90
101	Kochhar	90
...		
202	Fay	20
205	Higgins	110
206	Gietz	110

DEPARTMENTS

DEPARTMENT_ID	DEPARTMENT_NAME	LOCATION_ID
10	Administration	1700
20	Marketing	1800
50	Shipping	1500
60	IT	1400
80	Sales	2500
90	Executive	1700
110	Accounting	1700
190	Contracting	1700

EMPLOYEE_ID	DEPARTMENT_ID	DEPARTMENT_NAME
200	10	Administration
201	20	Marketing
202	20	Marketing
...		
102	90	Executive
205	110	Accounting
206	110	Accounting

Generating a Cartesian Product

```
SELECT last_name, department_name dept_name
FROM emps, depts;
```

Retrieving Records with Equijoins

```
SELECT e.employee_id, e.last_name, e.department_id,
d.department_id, d.location_id
FROM emps e , depts d
```

```
WHERE e.department_id = d.department_id;
```

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_ID	LOCATION_ID
200	Whalen	10	10	1700
201	Hartstein	20	20	1800
202	Fay	20	20	1800
124	Mourgos	50	50	1500
141	Rajs	50	50	1500
142	Davies	50	50	1500
143	Matos	50	50	1500

Creating Joins with the ON Clause

```
SELECT e.employee_id, e.last_name, e.department_id,
d.department_id, d.location_id
FROM emps e JOIN depts d
ON (e.department_id = d.department_id);
```

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_ID	LOCATION_ID
200	Whalen	10	10	1700
201	Hartstein	20	20	1800
202	Fay	20	20	1800
124	Mourgos	50	50	1500
141	Rajs	50	50	1500
142	Davies	50	50	1500
143	Matos	50	50	1500

Joining a Table to Itself

```
SELECT worker.last_name, "works for", manager.last_name
     FROM emps worker, emps manager
WHERE worker.Manager_id = manager.Employee_Id;
```

WORKER.LAST_NAME||'WORKSFOR'||MANAGER.LAST_NAME
Kochhar works for King
De Haan works for King
Mourgos works for King
Zlotkey works for King
Hartstein works for King
Whalen works for Kochhar
Higgins works for Kochhar
Hunold works for De Haan
Ernst works for Hunold

Topics:

After completing this lesson, you should be able to do:

- Creating Three-Way Joins with the ON Clause
- LEFT OUTER JOIN
- RIGHT OUTER JOIN
- FULL OUTER JOIN
- Additional Conditions

Creating Three-Way Joins with the ON Clause

```
SELECT employee_id, city, department_name
FROM emps e
JOIN depts d
ON d.department_id = e.department_id
JOIN locs l
ON d.location_id = l.location_id;
```

EMPLOYEE_ID	CITY	DEPARTMENT_NAME
103	Southlake	IT
104	Southlake	IT
107	Southlake	IT
124	South San Francisco	Shipping
141	South San Francisco	Shipping
142	South San Francisco	Shipping

LEFT OUTER JOIN

```
SELECT e.last_name, e.department_id, d.department_name
FROM emps e
LEFT OUTER JOIN depts d
ON (e.department_id = d.department_id) ;
```

LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
Whalen	10	Administration
Fay	20	Marketing
Hartstein	20	Marketing

....

De Haan	90	Executive
Kochhar	90	Executive
King	90	Executive
Gietz	110	Accounting
Higgins	110	Accounting
Grant		

RIGHT OUTER JOIN

```
SELECT e.last_name, e.department_id, d.department_name
FROM emps e
RIGHT OUTER JOIN depts d
ON (e.department_id = d.department_id) ;
```

LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
King	90	Executive
Kochhar	90	Executive

....

Whalen	10	Administration
Hartstein	20	Marketing
Fay	20	Marketing
Higgins	110	Accounting
Gietz	110	Accounting
		Contracting

FULL OUTER JOIN

```
SELECT e.last_name, e.department_id, d.department_name
FROM emps e
LEFT JOIN depts d ON e.department_id = d.department_id
UNION
SELECT e.last_name, e.department_id, d.department_name
FROM emps e
RIGHT JOIN depts d ON e.department_id = d.department_id;
```

LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME
Whalen	10	Administration
Fay	20	Marketing

...

De Haan	90	Executive
Kochhar	90	Executive
King	90	Executive
Gietz	110	Accounting
Higgins	110	Accounting
Grant		
		Contracting

Additional Conditions

```
SELECT e.employee_id, e.last_name, e.department_id,
d.department_id, d.location_id
FROM emps e JOIN depts d
ON (e.department_id = d.department_id)
AND e.manager_id = 149 ;
```

EMPLOYEE_ID	LAST_NAME	DEPARTMENT_ID	DEPARTMENT_ID	LOCATION_ID
174	Abel	80	80	2500
176	Taylor	80	80	2500

Activity 01:

Write a query to display the employee's last name, department name, location ID, and city of all employees who earn a commission.

Activity 02:

Write a query to display the last name, job, department number, and department name for all employees who work in Toronto.

LAST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME
Hartstein	MK_MAN	20	Marketing
Fay	MK_REP	20	Marketing

Activity 03:

Display the employee last name and employee number along with their manager's last name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.